READING PASSAGE 1

*You should spend about 20 minutes on **Questions 1–13**, which are based on Reading Passage 1 below.*

Learning by example: evidence from studies of rats and birds

The best known model of animal learning behaviour is probably the so-called classical model associated with the Russian scientist Pavlov. The key features of this model are stimulus, outcome and repetition. Pavlov demonstrated that animals could be trained to associate a sensory stimulus with a particular outcome, even where no natural association exists. In one famous experiment, dogs learned to associate the sound of a bell with the expectation of food. Subsequent models, such as the 'operant conditioning' model, also involved observing animals in isolation. Recently, however, behavioural research has focused on how animals are equipped to learn in social settings. And interest in the concept of role modelling, already a popular notion for guiding child development, has spread to other species.

Heyes and Galef (1996) have compiled some very interesting and accessible papers on social learning in mammals. One of these papers starts with an account of a school field trip to Israel. In a pine forest there, many pine cones were discovered which had been stripped to the central core. So an investigation was started which had no weighty theoretical intent, but was directed initially at finding out what were eating the nutritious pine seeds, and how they managed to get them out of the cones. The culprit proved to be the versatile and athletic black rat. The forest-dwelling rat used a standard technique, biting of each cone scale in sequence, from base to tip, following the spiral growth pattern of the cone.

The investigation was then broadened, with the aim of determining how the rat acquires such a skill. Urban black rats were found to lack the skill, and were unable to learn it even when they were housed with experienced cone strippers, such as those found in Israel. However, infants of urban mothers who were fostered to forest mothers acquired the skill, whereas infants of forest mothers fostered by urban mothers did not. Clearly, the skill is not inherited, but has to be taught.

Further elegant experiments showed that naive adult urban rats can develop the skill if they are provided with cones from which the first complete spiral of scales have been removed, rather as people can deduce how to use a new photocopier once they have been shown how to switch it on. In the case of forest rats, mothers supply their infants with cones which they have partially stripped, thereby providing the key for acquiring the complete stripping skill.

We might conclude that this is a good example of adaptive learning, but how economical is it? The researchers went on to determine this by measuring the oxygen uptake of a rat stripping a cone. The rat was placed in a metabolic chamber so that the amount of energy used in the cone-stripping operation could be calculated. This was then compared with the energy benefit of the pine seeds, as measured by a machine known as a calorimeter. The rat's energy expenditure proved to be less than 10% of the energetic value of the cone – an acceptable profit margin.

A paper by Bednekoff and Balda (1996) provides another example of the adaptiveness of social learning, this time in birds. It concerns the seed caching behaviour of two birds: Clark's Nutcracker (*Nucifraga columbiana*) and the Mexican Jay (*Aphelocoma ultramarina*). The former is a specialist, caching 30,000 or so seeds in scattered locations, from where it recovers them over the winter months. The Mexican Jay also caches food, but is not nearly as dependent on this as the Nutcracker. The two species also differ in their social structure, the Nutcracker being rather solitary, while the Jay forages in social groups.

An experiment was conducted to discover not just whether a bird can remember where it hid a seed itself, but also if it can remember where it saw another bird hide a seed. The design of the experiment is slightly comical, with a cacher bird wandering about a room with lots of holes in the floor and hiding food in some of them, while watched by an observer bird perched in a cage. Two days later, cachers and observers are tested for their discovery rate against an estimated random performance. In the role of cacher, not only the Nutcracker but also the less specialised Jay performed above the rate of chance. More surprisingly, however, Jay observers were as successful as Jay cachers, whereas Nutcracker observers did no better than chance. So it seems that, whereas the Nutcracker is highly adapted for remembering where it hid its own seed, the gregarious Mexican Jay is more adept than the Nutcracker at remembering, and so exploiting, the caches of other birds.

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caching: storing

Questions 1–4

*Reading Passage 1 has seven paragraphs **A–G**.*

Which paragraph contains the following information?

*Write the correct letter, **A–G**, in boxes 1–4 on your answer sheet.*

1 a comparison between learning in rats and learning in humans

2 an explanation of what initially prompted a research project

3 an overview of research into learning behaviour in animals

4 an analysis of cost-effectiveness

Questions 5–9

Do the following statements agree with the information given in Reading Passage 1?

In boxes 5–9 on your answer sheet, write

TRUE	*if the statement agrees with the information*
FALSE	*if the statement contradicts the information*
NOT GIVEN	*if there is no information on this*

5 The purpose of the field trip to Israel was to investigate the feeding habits of the black rat.

6 Pine cones are the main source of food for forest rats in Israel.

7 The forest rats in Israel had a fixed method of peeling pine cones.

8 Forest rats know instinctively how to peel pine cones.

9 In extracting pine seeds from the cones, rats use only a small proportion of the energy they gain from eating them.

Questions 10–13

*Complete the summary using the list of words, **A–J**, below.*

*Write the correct letter, **A–J**, in boxes 10–13 on your answer sheet.*

Clark's Nutcracker and the Mexican Jay are two species of bird which store seeds in various places for use in winter. The Jay relies **10** _____ on this habit, and is more **11** _____ in its feeding habits. During an experiment, one bird stored seeds while another **12** _____. Of the two species, the **13** _____ was consistently more able to find seeds which had been stored by other birds.

A	equally	**B**	Nutcracker	**C**	dependent	**D**	watched
E	solitary	**F**	observer	**G**	less	**H**	social
I	Jay	**J**	more				

Learning by example

段落信息匹配 | Questions 1-4

答案 1 D / 2 B / 3 A / 4 E

题 / 答案	题干与翻译	详细定位句与翻译	对应逻辑与易错点
1 **D**	**a comparison between learning in rats and learning in humans** 译：对老鼠学习与人类学习的比较。	**D段（第4段）** Further elegant experiments showed that naive adult urban rats can develop the skill if they are provided with cones from which the first complete spiral of scales have been removed, rather as people can deduce how to use a new photocopier once they have been shown how to switch it on. 译：进一步的巧妙实验表明：如果给没有相关经验的成年城市黑鼠提供已经剥掉完整第一圈鳞片的松果，它们就能学会这项技能；这很像人们一旦学会如何启动一台新复印机，便能推断出如何使用它。 In the case of forest rats, mothers supply their infants with cones which they have partially stripped, thereby providing the key for acquiring the complete stripping skill. 译：森林黑鼠的母鼠会把已经剥去一部分鳞片的松果提供给幼鼠，从而为幼鼠掌握完整的剥取技能提供关键提示。	**rather as 引出类比：老鼠与人类都能借助起步提示，进一步掌握完整操作。** 老鼠得到已剥掉第一圈鳞片的松果，由此学会继续剥取；人类先学会启动复印机，由此推断后续使用方法。两个学习过程被并列比较，直接对应 comparison between learning in rats and learning in humans。 C段比较城市黑鼠与森林黑鼠，没有涉及人类学习；A段提到儿童发展，只交代研究兴趣的扩展。题干要求两个对象的学习过程形成明确比较，D段的复印机类比最直接。
2 **B**	**an explanation of what initially prompted a research project** 译：解释最初是什么促成了一项研究。	**B段（第2段）** One of these papers starts with an account of a school field trip to Israel. 译：其中一篇论文首先记述了一次学校组织的以色列实地考察。 In a pine forest there, many pine cones were discovered which had been stripped to the central core. 译：人们在当地一片松林中发现了许多已经被剥得只剩中央轴芯的松果。 So an investigation was started which had no weighty theoretical intent, but was directed initially at finding out what were eating the nutritious pine seeds, and how they managed to get them out of the cones. 译：于是，一项调查开始了。它没有宏大的理论目标，最初只是想弄清哪些动物在吃富有营养的松子，以及它们如何从松果中取出松子。	**发现被剥空的松果→ 想弄清谁在吃、如何取出→ 启动调查。So 明确连接了发现与研究启动。** prompt 在这里作动词，表示促使、引发；initially 限定最初阶段。题干要找的是研究的起因，B段交代了直接触发调查的观察。 C段的 investigation was then broadened 表示研究随后扩大，研究者进一步追问黑鼠如何学会这项技能。这个后续研究目标不能替代最初的启动原因。
3 **A**	**an overview of research into learning behaviour in animals** 译：对动物学习行为研究的总体介绍。	**A段（第1段）** The best known model of animal learning behaviour is probably the so-called classical model associated with the Russian scientist Pavlov. 译：最广为人知的动物学习行为模型，可能是与俄罗斯科学家巴甫洛夫有关的所谓经典模型。 Subsequent models, such as the 'operant conditioning' model, also involved observing animals in isolation. 译：后来的模型，例如“操作性条件作用”模型，也涉及对处于隔离状态的动物进行观察。 Recently, however, behavioural research has focused on how animals are equipped to learn in social settings. 译：然而，近来的行为研究已转向动物如何具备在社会环境中学习的能力。 And interest in the concept of role modelling, already a popular notion for guiding child development, has spread to other species. 译：榜样示范这一概念原本已广泛用于指导儿童发展，如今研究者对它的兴趣也扩展到了其他物种。	**A段按研究的发展顺序，概括了不同模型与研究方向，因此属于 overview（总体介绍）。** 脉络为：巴甫洛夫的经典模型→ 后来的操作性条件作用模型→ 最近对社会情境中学习的关注。Subsequent 与 Recently 清楚标示研究的发展。 overview 不要求列出所有研究，而是从较大范围介绍这个领域的主要背景和变化。B-G段主要展开黑鼠或鸟类的具体研究，缺少A段这种跨模型的概览。
4 **E**	**an analysis of cost-effectiveness** 译：对成本效益的分析，即考察付出的成本是否值得。	**E段（第5段）** The rat was placed in a metabolic chamber so that the amount of energy used in the cone-stripping operation could be calculated. 译：黑鼠被放入代谢测量室，以便计算它剥松果过程中消耗的能量。 This was then compared with the energy benefit of the pine seeds, as measured by a machine known as a calorimeter. 译：随后，研究者将这一能量消耗与松子提供的能量收益作比较；后者由一种称为量热计的仪器测定。 The rat's energy expenditure proved to be less than 10% of the energetic value of the cone - an acceptable profit margin. 译：结果表明，黑鼠消耗的能量不到松果能量价值的10%，这样的收益幅度是可以接受的。	**cost（成本）对应剥取松子的能量消耗；benefit（收益）对应松子提供的能量。研究者将两者比较，判断这种取食方式是否划算。** This 指前一句计算出的能量消耗。代谢测量室用于评估消耗，量热计测量食物的能量收益；less than 10% 是两者比较后的结果。 因此，cost-effectiveness 在本文用能量投入和回报来衡量。能学会某种技能说明学习有效；把付出和收益放在一起比较，才对应本题要求的成本效益分析。

文章结构速览｜按正文七个自然段依次对应 A-G

A	Research background	学习模型与研究方向的变化。
B	The investigation begins	发现剥空的松果，追查动物身份及剥取方式。
C	Cross-fostering evidence	交叉抚养实验表明技能需要后天习得。
D	Learning from a starting cue	部分剥好的松果提供起步提示，并类比人类使用复印机。
E	Energy costs and benefits	比较剥松果的能量消耗与松子的能量收益。
F	Two bird species	介绍两种鸟的藏食习惯、依赖程度与社会结构。
G	Caching and observing	比较藏食者与观察者找回食物的表现。

Learning by example

事实判断 | Questions 5-9

答案 5 FALSE / 6 NOT GIVEN / 7 TRUE / 8 FALSE / 9 TRUE

题 / 答案	题干与翻译	详细定位句与翻译	判断依据与易错点
5 **FALSE**	**The purpose of the field trip to Israel was to investigate the feeding habits of the black rat.** 译：这次以色列实地考察的目的是调查黑鼠的进食习惯。	**B段（第2段）** One of these papers starts with an account of a school field trip to Israel. 译：其中一篇论文首先记述了一次学校组织的以色列实地考察。 In a pine forest there, many pine cones were discovered which had been stripped to the central core. 译：在当地一片松林中，人们发现了许多已经被剥得只剩中央轴芯的松果。 So an investigation was started which had no weighty theoretical intent, but was directed initially at finding out what were eating the nutritious pine seeds, and how they managed to get them out of the cones. 译：于是，一项调查开始了。它没有宏大的理论目标，最初只是想弄清哪些动物在吃富有营养的松子，以及它们如何把松子取出来。 The culprit proved to be the versatile and athletic black rat. 译：调查发现，干这件事的是适应能力强、行动敏捷的黑鼠。	**原文的事件顺序是：学校考察→ 发现被剥空的松果→ 因此启动调查→ 查明是黑鼠所为。** 题干将考察途中由发现引发的调查，写成了这次考察原本的目的，改变了事件之间的关系，因此判FALSE。So an investigation was started 给出了调查启动的直接原因。 原文没有交代学校考察的完整原定安排，但已经说明针对取食者的调查如何产生；而且调查开始时，研究者还在追问“是什么动物在吃”。这些信息足以反驳题干所指认的特定目的。
6 **NOT GIVEN**	**Pine cones are the main source of food for forest rats in Israel.** 译：松果是以色列森林黑鼠的主要食物来源。	**B段（第2段）** So an investigation was started which had no weighty theoretical intent, but was directed initially at finding out what were eating the nutritious pine seeds, and how they managed to get them out of the cones. 译：于是，一项调查开始了。它没有宏大的理论目标，最初只是想弄清哪些动物在吃富有营养的松子，以及它们如何把松子取出来。 The culprit proved to be the versatile and athletic black rat. 译：调查发现，干这件事的是适应能力强、行动敏捷的黑鼠。	**原文能够确认黑鼠从松果中取出松子食用，但没有说明这种食物在它们全部食物中占多大比例。** main source 要求“主要来源”这一地位；知道它们会吃某种食物，仍不足以判断它是不是最主要的来源。需要进一步的食物构成、频率或占比信息，文中没有提供。 松果能量收益较高，也不能直接推出黑鼠主要靠它生活。题干中的 pine cones 可理解为提供松子的食物来源；本题信息缺口在 main，故判NOT GIVEN。
7 **TRUE**	**The forest rats in Israel had a fixed method of peeling pine cones.** 译：以色列森林黑鼠剥松果时采用固定的方法。	**B段（第2段）** The forest-dwelling rat used a standard technique, biting off each cone scale in sequence, from base to tip, following the spiral growth pattern of the cone. 译：生活在森林中的黑鼠采用一种固定的技术：顺着松果的螺旋生长方向，从底部到顶部，依次咬掉每一片鳞片。	**a standard technique 直接对应 a fixed method，都是说有一套固定的操作方式。** 后面的三个细节将“固定”具体化：in sequence 指按顺序；from base to tip 指从底部到顶部；following the spiral growth pattern 指沿着螺旋结构进行。动作的顺序、方向和路径都有规则，故判TRUE。 peeling pine cones 概括了咬掉鳞片、剥开松果的操作。判断依据是这套动作关系，题干无需重复原文的每一个动作词。
8 **FALSE**	**Forest rats know instinctively how to peel pine cones.** 译：森林黑鼠凭本能就知道如何剥松果。	**C段（第3段）** However, infants of urban mothers who were fostered to forest mothers acquired the skill, whereas infants of forest mothers fostered by urban mothers did not. 译：然而，城市母鼠的幼鼠交由森林母鼠抚养后，学会了这项技能；森林母鼠的幼鼠交由城市母鼠抚养后，却没有学会。 Clearly, the skill is not inherited, but has to be taught. 译：显然，这项技能并不靠遗传获得，需要通过教导习得。	**instinctively 表示凭本能、无需后天学习。原文明确指出这项技能需要习得，与题干相反，故判FALSE。** 交叉抚养提供了两组证据：城市幼鼠由森林母鼠抚养→ 学会；森林幼鼠由城市母鼠抚养→ 没学会。结果随抚养环境变化，出生于森林母鼠本身并不足以保证掌握技能。 D段进一步解释，森林母鼠提供部分剥好的松果，帮助幼鼠掌握完整操作。不能由“森林鼠会剥松果”直接推出“它们天生就会”。
9 **TRUE**	**In extracting pine seeds from the cones, rats use only a small proportion of the energy they gain from eating them.** 译：黑鼠从松果中取出松子时消耗的能量，只占它们食用松子所得能量的一小部分。	**E段（第5段）** The rat was placed in a metabolic chamber so that the amount of energy used in the cone-stripping operation could be calculated. 译：黑鼠被放入代谢测量室，以便计算剥松果这一操作所消耗的能量。 This was then compared with the energy benefit of the pine seeds, as measured by a machine known as a calorimeter. 译：随后，研究者将这一消耗与松子提供的能量收益作比较，后者由量热计测定。 The rat's energy expenditure proved to be less than 10% of the energetic value of the cone - an acceptable profit margin. 译：结果表明，黑鼠消耗的能量不到松果能量价值的10%，这样的收益幅度是可以接受的。	**less than 10% 对应 only a small proportion，原文的精确比例被题干概括为“一小部分”，因此判TRUE。** 比较的两项是“剥取松子消耗的能量”和“吃松子获得的能量”。例如，若松子能提供100单位能量，剥取它们消耗的能量就不到10单位；这个例子只用于说明比例。 这里的 energy expenditure 由前文限定为 cone-stripping operation 的消耗，不能扩大为黑鼠一天的全部能量消耗。题干中的 eating them 指食用松子。

Learning by example

选词摘要填空 | Questions 10-13

答案 10 G (less) / 11 H (social) / 12 D (watched) / 13 I (Jay)

题 / 答案	摘要题干与翻译	详细定位句与翻译	结构、对应与排除
10 **G**	**The Jay relies 10 _____ on this habit,...** 译：松鸦对这种习惯的依赖程度_____。 答案：less（较少地）。 this habit 指前文提到的储藏种子、留待冬季食用的习惯。	**F段（第6段）** The former is a specialist, caching 30,000 or so seeds in scattered locations, from where it recovers them over the winter months. 译：前者擅长储藏食物，会把大约三万颗种子藏在分散的地点，再在冬季从这些地点取回。 The Mexican Jay also caches food, but is not nearly as dependent on this as the Nutcracker. 译：墨西哥松鸦也储藏食物，但它对这种行为的依赖程度远不如星鸦。	**not nearly as dependent... as 表示“依赖程度远不如……”，所以松鸦对储食习惯的依赖较少，选G：less。** 填回后：The Jay relies less on this habit. rely on 是“依赖”，less 修饰依赖的程度。the former 指前文先提到的 Clark's Nutcracker（克拉克星鸦）。 A equally 表示同样程度，J more 表示更多，均与比较方向不符。C dependent 是形容词，适合 be dependent on；放入 relies dependent on 不成立。
11 **H**	**and is more 11 _____ in its feeding habits.** 译：并且在取食习惯上更具有_____性。 答案：social（群体的、社会性的）。 完整句意：松鸦取食时更倾向于群体活动。	**F段（第6段）** The two species also differ in their social structure, the Nutcracker being rather solitary, while the Jay forages in social groups. 译：这两个物种的社会结构也不同：星鸦比较独居，而松鸦会成群觅食。	**forages in social groups 表示“成群觅食”，正好对应取食习惯更具 social（群体性）的描述，选H。** 结构为 is more + 形容词。social 可以放入这个位置；forage 指寻找食物，与题干 feeding habits 属于同一取食情境。 E solitary（独居的）在原文描述的是星鸦，换给松鸦会颠倒对象。C dependent（依赖的）涉及储食依赖程度，前文还明确说松鸦依赖更少，无法支持 is more dependent。
12 **D**	**During an experiment, one bird stored seeds while another 12 _____.** 译：在一项实验中，一只鸟储藏种子，同时另一只鸟_____。 答案：watched（在旁观察）。	**G段（第7段）** The design of the experiment is slightly comical, with a cacher bird wandering about a room with lots of holes in the floor and hiding food in some of them, while watched by an observer bird perched in a cage. 译：实验的设计略显滑稽：一只负责藏食的鸟在地板布满洞孔的房间里走动，把食物藏进其中一些洞里；同时，一只栖在笼中的观察鸟注视着它。	**两只鸟的角色是 cacher（藏食者）和 observer（观察者）。前者藏种子，后者观看，故选D：watched。** 原文 while watched by an observer bird 从藏食者角度采用被动表达；题干 while another watched 将观察鸟作为主语，改成主动表达。观察关系保持一致。 another 在这里代替 another bird，已经充当主语，后面需要动词与 stored 对应。F observer 是“观察者”这个名词；another observer 无法构成完整的 while 从句。
13 **I**	**Of the two species, the 13 _____ was consistently more able to find seeds which had been stored by other birds.** 译：在这两个物种中，_____始终更擅长找到其他鸟储藏的种子。 答案：Jay（松鸦）。 注意限定：stored by other birds，即“由其他鸟储藏”。	**G段（第7段）** In the role of cacher, not only the Nutcracker but also the less specialised Jay performed above the rate of chance. 译：在藏食者这一角色中，星鸦和较不专门擅长藏食的松鸦都表现得好于随机水平。 More surprisingly, however, Jay observers were as successful as Jay cachers, whereas Nutcracker observers did no better than chance. 译：然而，更令人惊讶的是，作为观察者的松鸦与作为藏食者的松鸦一样成功，而作为观察者的星鸦，其表现没有好于随机水平。 So it seems that, whereas the Nutcracker is highly adapted for remembering where it hid its own seed, the gregarious Mexican Jay is more adept than the Nutcracker at remembering, and so exploiting, the caches of other birds. 译：因此看来，星鸦很擅长记住自己藏种子的地点；而合群的墨西哥松鸦比星鸦更擅长记住并利用其他鸟的藏食地点。	**题干问“找回其他鸟藏的种子”，应比较观察者的表现：松鸦观察者表现良好，星鸦观察者只达到随机水平，所以选I：Jay。** B Nutcracker 容易被 specialist 和三万颗种子的储藏量吸引，但这些信息侧重储藏及记住自己的食物。自己藏的位置与看见别的鸟藏的位置，是实验分别考查的两种能力。 Of the two species 要求填写物种名称，F observer 只是实验角色。did no better than chance 表示没有超过随机找到的水平，不能理解为一颗种子也找不到。

选项 A-J｜答案栏填写字母

A	equally	同等地；程度相同。
B	Nutcracker	星鸦；本文指克拉克星鸦。
C	dependent	依赖的。
D	watched	观察；watch 的过去式。
E	solitary	独居的；惯于单独活动的。
F	observer	观察者；名词。
G	less	较少地；程度更低。
H	social	群体的；社会性的。
I	Jay	松鸦；本文指墨西哥松鸦。
J	more	更多地；程度更高。